

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			Half-life is <u>too</u> short / <u>only</u> 13 hours (1) Don't accept very short or shorter So will decay [too] quickly [to treat cancer] / will need replacing frequently / doesn't last long enough [to treat the cancer] (1)		2		2		
	(b)			Half-life is longer / decays more slowly (1) [Combination of beta and] gamma will be more penetrating (1)		2		2		
	(c)			80 days = 10 half-lives (1) Series of halving starting from 1 or 100 % or 1 000 e.g. $1 \rightarrow \frac{1}{2} \rightarrow \frac{1}{4} \rightarrow \frac{1}{8}$ etc (1) Answer = $\frac{1}{1024}$ so 80 days is long enough or it will not trigger the monitors (1)			3	3	3	
				Question 10 total	0	4	3	7	3	0